

Code: PersonManager.java

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import java.io.*;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;
import java.util.List;

public class PersonManager {
    private List<Person> persons = new ArrayList<>();

    public void addPerson(Person person) {
        persons.add(person);
    }

    public void updatePerson(int id, String name, int age, String address) {
        for (Person person : persons) {
            if (person.getId() == id) {
                person.setName(name);
                person.setAge(age);
                person.setAddress(address);
                break;
            }
        }
    }

    public void deletePerson(int id) {
        for (Person person : persons) {
            if (person.getId() == id) {
                persons.remove(person);
                break;
            }
        }
    }

    public boolean checkConstraint(int id) {
        for (Person person : persons) {
            if (person.getId() == id) {
                return false;
            }
        }
        return true;
    }

    public void sortByName() {
        Collections.sort(persons, Comparator.comparing(Person::getName));
    }

    public List<Person> searchByName(String name) {
        List<Person> result = new ArrayList<>();
        for (Person person : persons) {
            if (person.getName().equals(name)) {
                result.add(person);
            }
        }
        return result;
    }

    public void saveToFile(String fileName) throws IOException {
        try (ObjectOutputStream oos = new ObjectOutputStream(new FileOutputStream(
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        fileName))) {
            oos.writeObject(persons);
        }
    }

    public void loadFromFile(String fileName) throws IOException,
        ClassNotFoundException {
        try (ObjectInputStream ois = new ObjectInputStream(new FileInputStream(
            fileName))) {
            persons = (List<Person>) ois.readObject();
        }
    }

    @Override
    public String toString() {
        StringBuilder sb = new StringBuilder();
        for (Person person : persons) {
            sb.append(person.toString()).append("\n");
        }
        return sb.toString();
    }
}

```